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Climate policy uncertainty and corporate trade credit provision

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ABSTRACT

This study examines how uncertainty in Chinese climate policies influences firms' trade credit decisions. Using data from Chinese A-share listed companies between 2010 and 2022, we find that increased climate policy uncertainty is associated with reduced trade credit provision. These findings remain robust across various endogeneity tests and additional verification methods. Our analysis reveals that financial constraints, operating risk, and information opacity are three plausible channels through which climate policy uncertainty to inhibit trade credit provision. Furthermore, cross-sectional analyses demonstrate that this negative relationship is more pronounced among non-state-owned enterprises, firms with weak political connections, and companies operating in less competitive product markets. Overall, this paper advances the academic discourse on trade credit determinants and provides new insights into how climate policy uncertainty affects corporate credit provision decisions.

1. Introduction

Climate change has led to various extreme weather events that significantly threaten individual wealth and financial markets in recent years. In response, authorities worldwide have made firm commitments to address climate change.[1] However, these ongoing policy adjustments have inevitably heightened climate policy uncertainty (henceforth, CPU),[2] which critically impacts financial markets (Hong et al., 2020). Prior research has documented that CPU affects various aspects of corporate behavior, including firm performance (Ongsakul et al., 2023), innovation activities (Bai et al., 2023; Sun et al., 2024), firm investment (Xu et al., 2024; Hu et al., 2023; Zhao et al., 2025), total factor productivity (Ren et al., 2022), corporate stock returns (He and Zhang, 2022; Treepongkaruna et al., 2023) and greenwashing behavior (Wang et al., 2026). While these studies have illuminated the economic impact of CPU, they predominantly focus on corporate internal-market outcomes or business operation decisions. Limited attention has been given to the effect of CPU on corporate financing, particularly trade credit financing.

Trade credit, which occurs when firms defer payments or receive prepayments, represents an essential form of external financing between firms (Petersen and Rajan, 1997). This financing mechanism affects approximately 80% of businesses globally (Cao et al., 2024). In emerging markets with underdeveloped financial systems, particularly China, trade credit has proven to act as an important

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[1] The 2015 Paris Climate Agreement, signed by 195 nations, established a commitment to limit global surface temperature increases to 2 °C above pre-industrial levels. Following this, the 2021 Glasgow Climate Pact garnered signatures from more than 140 countries, with signatories pledging to achieve net-zero emissions.

[2] While the Paris Agreement established a coordinated framework for international climate action, its implementation has generated considerable policy uncertainty worldwide. The agreement's structure relies on nationally determined contributions, requiring countries to periodically update their commitments and leading to frequent policy adjustments among participants. Despite these efforts, the persistent gap between actual emissions trajectories and climate targets has exposed implementation difficulties and necessitated increasingly ambitious policy interventions (Allen et al., 2019). Moreover, policy reversals such as the U.S. withdrawal process have further intensified international coordination challenges and policy uncertainty.

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substitute for bank financing and provides vital short-term capital for enterprises (Klapper et al., 2011). Given the critical role of trade credit, shocks that trigger changes in trade credit supply may have profound implications for economic development.

Previous research shows that trade credit serves as a financing mechanism proactively extended by suppliers to stimulate commercial sales and services, and has identified various determinants of this practice. Specifically, factors such as firm-level commercial guarantees (Safiullah and Nguyen, 2024), culture (Li et al., 2020), and competitive position can influence information opacity on both supply and demand sides, thereby affecting trade credit (Kong et al., 2020; Fisman and Raturi, 2004). Other studies by Song and Li (2024) and Barrot (2016) find that property rights and corporate governance structures are closely related to financing conditions, which in turn affect trade credit dynamics (Ferrando and Mulier, 2013). Furthermore, following Love et al. (2007)'s seminal work on the macroeconomic shock of financial crises on trade credit, recent studies have also expanded to examine uncertainty impacts, including economic policy uncertainty and monetary policy fluctuations (Jory et al., 2020; Adelino et al., 2022). While this research does not directly examine the effect of CPU on trade credit, it provides a theoretical foundation for investigating this relationship in our study.

CPU is expected to reduce firms' trade credit provision for at least three reasons: The first two channels emphasize that short-term adverse shocks, including policy changes, can trigger inefficient fluctuations in both financing and operations, potentially increasing risk. Specifically, the first channel highlights that CPU increases the insolvency risks borne by banks and discourages their lending, as found by Dai and Zhang (2023) and Golub et al. (2020). This deterioration of financing conditions adversely affects suppliers' own borrowing terms, prompting them to adopt more conservative trade credit supply strategies (Love and Zaidi, 2010; Cosci et al., 2019). The second channel focuses on the disturbance of corporate operations triggered by CPU. This resultant curtailment of firms' labor demand, production capacity, and investment returns can increase their risk exposure, as stressed by Ren et al. (2022) and Huang and Sun (2023). In response, suppliers may preemptively reduce accounts receivable to minimize their exposure to potential financial distress. The third channel emphasizes that CPU intensifies information asymmetry between firms and their economic stakeholders. This leads to heightened risk of purchase returns, elevates order cancellation rates, and incurs burdensome decision costs in supplier-customer relationships. Consequently, these factors undermine firms' willingness to extend trade credit, as found by Ren et al. (2022) and Huang and Sun (2023).

In contrast, trade credit may become more attractive to firms during periods of heightened CPU. First, when access to traditional funding sources becomes limited, trade credit can serve as a vital alternative financing channel for small and medium-sized enterprises (Petersen and Rajan, 1997). In such cases, suppliers may be strongly motivated to proactively offer deferred payment terms to safeguard the financial lifeline of these constrained firms (Xiaojun Shi and Tan, 2020; Bussoli et al., 2023). Second, if CPU negatively disrupts corporate production, supplier firms may strengthen existing supply chain relationships by offering more trade credit to secure the stability of future orders (Daripa and Nilsen, 2011). Building on the preliminary theoretical analysis discussed above, the relationship between CPU and corporate trade credit provision yields competing predictions. Therefore, the nature of this relationship remains an empirical one.

To investigate this relationship as discussed above, we analyze data from Chinese A-share listed companies during 2010–2022. The reasons for focusing on the Chinese market are as follows. First, China has been the world's largest carbon emitter since 2006, according to the International Energy Agency. As of 2022, China's annual carbon emissions reached approximately 11.5 billion tons, constituting one-third of global carbon emissions. Therefore, Chinese climate policy changes are critical for addressing global climate challenges and meeting international emission reduction targets.

Additionally, China is actively addressing climate change. Beyond signing international agreements like the Paris Agreement, China has set self-imposed targets, including achieving peak carbon emissions by 2030 and reaching carbon neutrality by 2060, while implementing extensive climate policies to fulfill its international commitments.[3] As a major energy consumer, these policy transitions substantially affect China's real economy and financial system. Since the Chinese government plays a central role in economic planning and policy implementation, the centralized policy-making system means that climate policy changes can generate broad and direct impacts across industries and regions. Local governments also serve as key intermediaries in policy execution, further amplifying uncertainty transmission mechanisms and creating cascading effects throughout the economy (Handley and Limão, 2017).

Finally, China's financial system, as an emerging capital market, remains underdeveloped compared to mature economies. The system is predominantly bank-dominated, with capital markets playing a limited role in corporate financing (Allen et al., 2005; He and Wei, 2023). Financial repression characteristics are evident, including interest rate controls, credit rationing, and extensive government intervention in credit allocation (Shao et al., 2015; Cong et al., 2019). Moreover, government policy guarantees for state-owned enterprises (SOEs) and banks' preference for lending to firms with strong political connections and collateral assets have led the banking system to disproportionately favor SOEs, forcing many non-state-owned enterprises (Non-SOEs) to rely heavily on alternative financing channels for working capital needs (Allen et al., 2017; Cull et al., 2009). Consequently, trade credit has become extensively utilized among Chinese firms, with interfirm credit volume in some years even exceeding bank loans obtained by enterprises (Allen et al., 2005; Ge and Qiu, 2007). This unique institutional setting, characterized by centralized policy-making and financial market segmentation, provides an ideal context for studying how CPU affects firms' trade credit provision decisions.

To empirically investigate this relationship, we obtain the Chinese Climate Policy Uncertainty index (henceforth, CCPU), developed by Ma et al. (2023). Our annual CCPU measure is computed as the equal-weighted average of monthly indices.

[3] For further policy details, readers are directed to the official document available at the Ministry of Ecology and Environment of the People's Republic of China website: https://www.mee.gov.cn/ywgz/ydqhbh/qhbhlf/202211/W020221111763716523691.pdf.

Following Liu and Wang (2023), we measure a firm's trade credit provision as the ratio of combined accounts receivable, notes receivable, and prepayments to total assets. Controlling for various trade credit determinants and firm fixed effects, our regression analysis reveals a negative relationship between CCPU and trade credit provision.

Our findings remain robust across alternative measures of the CCPU index and trade credit, as well as subsample regression analyses. Although firm fixed effects in our baseline model mitigate most endogeneity concerns, this paper employs a two-stage least squares approach to strengthen causal interpretation. We then evaluate three potential channels through which CPU reduces trade credit provision: increased corporate financing constraints, information opacity, and operational risks. Finally, our cross-sectional heterogeneity analysis shows that the negative relationship between CPU and trade credit provision is more pronounced among non-state-owned enterprises and firms characterized by low industry competition and limited political connections.

This study makes three primary contributions. First, it extends the burgeoning body of research on CPU by examining its consequences from the perspective of firms' trade credit policies. While prior studies have documented CPU's negative effects on various firm activities (e.g., Pankratz et al. 2023, Bai et al. 2023, Bogmans et al. 2024) and its impacts on financial markets, including commodity market financialization (Ren et al., 2025), energy market connectedness (Yan and Cheung, 2025), and stock market returns (Chiang, 2025), we contribute by examining CPU's effect on alternative financing channels. Specifically, our study reveals an additional economic consequence of CPU—the reduction in trade credit provision, which serves as a crucial financing source for firms.

Second, we contribute to the literature examining how uncertainty shocks affect corporate trade credit. Previous research has examined various types of shocks, including economic policy (D'Mello and Toscano, 2020; Jory et al., 2020), monetary policy (Gertler and Karadi, 2015; Adelino et al., 2022) and financial crisis (Love and Zaidi, 2010). Our paper extends this literature by introducing the role of CPU and documenting its adverse effect on firm trade credit.

Third, our empirical model accounts for the possibility that climate policy changes may introduce negative externalities for corporate financing conditions, operating risks, and market information transparency—factors that are crucial to trade credit provision. Given that climate change represents a global challenge prompting policy adjustments across developed and emerging economies, and that trade credit serves as a prevalent interfirm financing mechanism worldwide, the transmission channels we identify possess broad theoretical relevance beyond China. These findings provide valuable insights for policymakers developing effective climate policies and help firms develop strategies to mitigate trade credit risks arising from CPU. Overall, our study provides a comprehensive understanding of the link between CPU and trade credit provision, with implications for other institutional contexts undergoing similar climate policy transitions.

The remainder of this paper is organized as follows. The next section briefly reviews the relevant literature and develops our hypotheses. Section 3 describes the sample, data, and empirical methodology, while Section 4 presents our empirical results. Finally, Section 5 concludes.

2. Literature review and hypotheses development

2.1. The determinants of trade credit provision

Trade credit is a widespread business practice and serves as a crucial source of external financing for firms. Prior literature provides diverse explanations for why firms extend trade credit. Firstly, according to the alternative financing theory proposed by Petersen and Rajan (1997) and Biais and Gollier (1997), credit rationing may restrict some borrowers' access to bank loans, even if they accept higher interest rates. As a result, these firms turn to their suppliers for credit. Suppliers derive informational benefits from their established business proximities with customers, leading to greater transparency. This informational advantage enables them to assess customer credit risk more accurately than banks. Moreover, monitoring inventories or fixed assets is often more reliable than evaluating cash flows, enabling suppliers to monitor these assets more effectively than banks (Burkart and Ellingsen, 2004). The combination of these supplier advantages and customer financing needs drives trade credit provision.

On the other hand, trade credit extension can be motivated by suppliers' business strategies. For example, when suppliers need to speed up the sale of goods, they may proactively provide trade credit (Giannetti and Koskinen, 2010). However, when enterprises are in urgent need of cash flow or face financing risk, the provision of trade credit will become conservative (Love et al., 2007). This evidence suggests that market stability is a prerequisite for suppliers to extend trade credit.

Overall, previous literature suggests that suppliers who are financially stable, face low operating risks, and maintain effective communication with downstream customers are more inclined to extend trade credit proactively. This study aims to explore whether and how suppliers modify their trade credit policies in response to CPU, which can both destabilize supply chain relationships and increase the marginal costs of trade credit provision.

2.2. Uncertainty and trade credit provision

Numerous studies have extensively examined suppliers' motivations for extending trade credit under stable policy conditions, including providing liquidity support, mitigating customers' financing constraints, and accelerating product sales (Fabbri and Menichini, 2010; Burkart and Ellingsen, 2004). However, external policy environments are rarely static. Especially in recent years, countries' frequent adjustments to climate policies in response to environmental risks have introduced significant uncertainty for businesses. Despite this trend, the relationship between CPU and firms' trade credit provision remains unexplored.

Recent studies increasingly emphasize the negative effects of uncertainty on trade credit. According to Gulen and Ion (2015), heightened uncertainty reduces firms' capital investment, which, in turn, affects financial stability and limits collateral assets, weakening companies' ability to secure credit from suppliers. Francis et al. (2014) posits that heightened policy uncertainty raises financing costs in the real sector, including trade credit. Supporting evidence from the supply side is also found in the work of Vo et al. (2022). Further, D'Mello and Toscano (2020) found that increased economic policy uncertainty leads to more volatile corporate cash flows, significantly higher default risk, and restricted commercial credit financing. More recently, Zhao et al. (2025) found that CPU significantly reduces corporate investment in China, suggesting that heightened CPU prompts firms to adopt more conservative financial strategies, which may extend to their trade credit provision decisions. These studies imply that suppliers, in response to uncertainty risks, may reduce their trade credit provision.

Our theoretical foundation is grounded in financial flexibility theory, which posits that firms maintain financial flexibility to respond effectively to unexpected shocks and seize profitable investment opportunities (Denis and Sibilkov, 2009). Trade credit provision, as a form of financial commitment that ties up working capital and reduces available liquidity, directly limits a firm's financial flexibility. When facing heightened uncertainty, firms have stronger incentives to preserve financial flexibility by holding larger cash reserves and reducing non-essential financial commitments. This framework is especially relevant in the context of CPU, where firms face elevated operational and financial risks that demand greater financial resilience.

Moreover, real options theory provides additional insights into this mechanism. Under uncertainty, the option value of "waiting" increases, making firms more reluctant to make irreversible financial commitments such as extending trade credit (Bloom et al., 2007; Jia and Li, 2020). The combination of these theoretical perspectives suggests that CPU should systematically reduce firms' willingness to provide trade credit.

We posit that CPU can reduce trade credit provision through three channels. First, funding liquidity constraints arise when frequent climate policy changes trigger liquidity shocks, forcing financially constrained firms to prioritize cash preservation over trade credit extension. Specifically, elevated CPU compromises firms' access to external financing, as climate-related risks complicate banks' risk assessment processes, making it more challenging to evaluate firms' repayment capacity accurately (Ginglinger and Moreau, 2023). Consequently, banks implement more stringent lending criteria, which increases external funding costs and restricts corporate financing. To maintain financial flexibility under these constraints, firms adopt more conservative approaches to extending trade credit. Francis et al. (2014) demonstrates that high CPU environments increase external financing costs, adversely affecting firms' future cash flows and destabilizing their financial position. This finding aligns with (Garcia-Appendini and Montoriol-Garriga, 2013), who demonstrate that financially constrained companies typically limit their trade credit extended to customers.

Second, operational risk management considerations lead firms to reduce trade credit supply when CPU creates uncertainty in corporate production and operations. Research demonstrates that CPU negatively impacts corporate operations, reducing production efficiency and market demand while creating ambiguity in future profitability and cash flow projections (Ren et al., 2022; Lee et al., 2021). Under these conditions, maintaining financial flexibility becomes crucial for operational stability. While trade credit typically serves as a short-term investment strategy to support clients and expand business opportunities, extending credit at the expense of liquidity may compromise firms' ability to respond to CPU-induced operational disruptions. High CPU can also lead to lower profits, financial distress, and heightened default risks (Dafermos et al., 2018; Lamperti et al., 2019), further reinforcing firms' incentives to maintain larger precautionary cash reserves rather than extending trade credit.

Third, information asymmetry and monitoring costs increase under CPU, eroding suppliers' traditional information advantage in trade credit provision. According to Petersen and Rajan (1997)'s lending advantage theory, suppliers gain cost efficiencies in acquiring customer information through close transactional relationships. However, CPU affects both the macroeconomic environment and supplier-customer transparency (Cao et al., 2024), with firms responding by engaging in selective disclosure to conceal negative impacts (Wang et al., 2026), thereby complicating suppliers' assessment of customer creditworthiness. The increased complexity of evaluating customer business sustainability and long-term profitability under uncertain climate policies may prompt suppliers to restrict credit or implement stricter lending standards to preserve their financial flexibility.

Hypothesis. Climate policy uncertainty is negatively associated with trade credit provision, ceteris paribus.

However, two counterarguments suggest that our hypothesized relationship may not hold, or that CPU might actually enhance trade credit provision. First, McGuinness and Hogan (2016) highlight that during macroeconomic credit market disruptions, firms often seek trade credit as an alternative financing source, resulting in increased demand. In such scenarios, extending additional trade credit can drive sales growth and expand market share. Second, Cunat (2007) found that during uncertain periods, firms often attract customers by offering more favorable trade credit terms, thereby mitigating the adverse effects of deteriorating macroeconomic conditions. Thus, companies may extend more trade credit during periods of heightened CPU.

3. Research design

3.1. Data and samples

We investigate the relation between CCPU and firms' trade credit provision in this paper. China issued its Climate Change Policies and Actions in 2009 and participated in the Copenhagen Climate Conference. Given the typical lagged effects of policy, 2010 marked the beginning of its substantive economic impact. Subsequently, spanning from 2011 with the introduction of carbon intensity targets in the 12th Five-Year Plan, through 2015 with participation in the Paris Agreement, to 2020 with the announcement of dual-carbon

goals, this period captured the most intensive stage of China's climate policy formulation and implementation. Hence, we construct our sample using firms listed on China's two major stock exchanges (Shanghai and Shenzhen) over the period 2010–2022, which provides sufficient temporal variation in CCPU while covering the entire policy cycle from initial international commitments to domestic implementation.

The financial statement data and trade credit information were obtained from the China Stock Market & Accounting Research (CSMAR) and WIND databases. Following Ren et al. (2022) and Xiu et al. (2023), the sample excludes: (1) firms under special treatment status (ST, ST*, PT), (2) financial institutions, (3) observations with missing or abnormal data. All continuous variables are winsorized at the 1st and 99th percentiles to minimize the impact of outliers. The final sample comprises 31,924 firm-year observations.

3.2. Measures of trade credit provision

This study investigates firms' trade credit provision, measured by the extent of credit sales that suppliers extend to their customers. Following Luo et al. (2023), Chen et al. (2019), and Garcia-Appendini and Montoriol-Garriga (2013), this paper measure trade credit provision (TC) as the ratio of the sum of accounts receivable, notes receivable, and prepayments to total assets. A higher TC value indicates greater trade credit extension. Chen et al. (2019) argues that accounts receivable represents the most direct form of trade credit extended to customers, while in emerging markets where alternative credit instruments are common, inclusion of notes receivable and prepayments is crucial for capturing the comprehensive supplier-customer financing relationship.

For robustness, we employ three alternative trade credit measures. The first two measures, TC_Rev and TC_Cor, are calculated as the sum of accounts receivable, notes receivable, and prepaid accounts, divided by sales and operating costs, respectively. Additionally, following Cao et al. (2024) and Jory et al. (2020), we consider net trade credit provision ($NetTC$). This measure is computed by subtracting accounts payable, notes payable, and prepaid accounts from the sum of accounts receivable, notes receivable, and prepaid accounts, and then dividing the result by total assets.

3.3. Measures of climate policy uncertainty

We use the CCPU index from Ma et al. (2023), which builds on the algorithmic model of Gavriilidis (2021). This index employs text mining and deep learning techniques to analyze climate policy news and related economic consequences in 1,755,826 articles from six major Chinese newspapers: People's Daily, Guangming Daily, Economic Daily, Global Times, Science and Technology Daily, and China News Service.

Gulen and Ion (2015) demonstrate that the explanatory power of uncertainty indices can be largely attributed to their news-based components, validating the effectiveness of newspaper-based uncertainty measures. Given that this index is constructed from major Chinese state-affiliated newspapers with significant policy influence, it provides an accurate measure of CPU. For our analysis, we employ annual data from this index and divide the original values by 100 to facilitate interpretation of the coefficient results.

3.4. Empirical model

We first examine the general effect of CCPU on firms' trade credit provision by estimating the baseline regression model in Eq. (1):

$$TC_{i,t+1} = \beta_0 + \beta_1 CCPU_t + \beta_2' \mathbb{M}_{i,t} + Firm_i + \varepsilon_{i,t} \tag{1}$$

where i and t index firm and year, respectively. The dependent variable, trade credit provision (TC), increases as firms provide more trade credit. Our key explanatory variable, $CCPU_t$, measures Chinese climate policy uncertainty. The vector $\mathbb{M}_{i,t}$ stacks all other control variables, including economic policy uncertainty (EPU), firm size ($Size$), firm age (Age), leverage (Lev), return on assets (ROA), sales growth ($Growth$), operation cash flow ratio ($Cash$), market-to-book ratio (BM), earnings before interest and tax ($EBIT$), largest shareholder's ownership ratio ($Top1$), and independent directors ratio ($Indep$) (Love et al., 2007; Luo et al., 2023; Zhou and Li, 2023; Cao et al., 2024). To mitigate potential reverse causality, we lag all independent variables by one year relative to the dependent variable. We incorporate firm fixed effects ($Firm_i$) to control for firm-specific characteristics. The model excludes year fixed effects because the $CCPU$ index is uniform across firms within each year, and their inclusion would result in perfect multicollinearity (Huang et al., 2022). We use heteroskedasticity-robust standard errors clustered at the firm level to ensure reliable statistical inference. The term $\varepsilon_{i,t}$ represents the unobserved exogenous error. By explicitly incorporating these factors, we ensure that the coefficient for $CCPU$ accurately reflects its incremental impact on trade credit provision. Our focal parameter, β_1, is expected to be negative. Complete variable definitions are presented in Table A.1.

4. Empirical results

4.1. Descriptive statistics

Table 1 presents descriptive statistics for variables used in the baseline regression. TC has a mean of 0.1659 and a standard deviation of 0.1178, with values ranging from 0.0030 to 0.5292, indicating substantial variation in trade credit provision across firms. $CCPU$ has a mean of 2.5769 and exhibits significant variation, with a range of 2.2127 to 3.1998, aligning with findings by Wu and Hu (2024).

Table 1
Descriptive statistics.

Variable	N	Mean	SD	Min	Median	Max
TC	31,924	0.1659	0.1178	0.0030	0.1469	0.5292
CCPU	31,924	2.5769	0.3279	2.2127	2.4476	3.1998
EPU	31,924	3.9495	2.3564	0.9889	3.6483	7.9187
Size	31,924	22.1449	1.2990	19.7736	21.9509	26.1636
Age	31,924	2.0224	0.9337	0.0000	2.1972	3.3322
Lev	31,924	0.4172	0.2090	0.0504	0.4073	0.9136
ROA	31,924	0.0403	0.0592	−0.2346	0.0395	0.1994
Growth	31,924	−0.3683	2.7927	−14.9294	0.0426	8.5689
Cash	31,924	0.0475	0.0684	−0.1612	0.0469	0.2430
Fix	31,924	0.2123	0.1592	0.0021	0.1798	0.6881
BM	31,924	0.6106	0.2497	0.1058	0.6089	1.1791
EBIT	31,924	0.6158	0.7793	−1.2584	0.4447	4.4629
Top1	31,924	34.6211	14.8817	8.5438	32.4379	74.2950
Indep	31,924	0.3755	0.0533	0.3333	0.3636	0.5714

Notes: This table presents descriptive statistics for variables used in the baseline regression. The sample comprises public firms in China covered by both CSMAR and Wind databases from 2010 to 2022. All variables are defined in detail in Table A.1.

4.2. The effect of CCPU on trade credit provision

We present the results estimating Eq. (1) in Table 2. Column (1) considers only CCPU as the independent variable, while column (2) incorporates EPU, and column (3) includes additional control variables. The CCPU coefficient estimates are consistently negative and statistically significant across all specifications, with significance levels remaining robust to the inclusion of control variables. This suggests that firms' trade credit provision decreases as CCPU increases, as we would have expected.

This negative effect is both statistically significant and economically substantial. Specifically, in column (3), coefficient of −0.0089 implies that a one standard deviation increase in CCPU reduces firms' trade credit provision by approximately 1.76 percentage points ((−0.0089*0.3279*100)/0.1659), assuming ceteris paribus during the study period.

4.3. Robustness checks

This section presents an array of robustness tests for our main findings. These include using alternative measures of trade credit provision and CCPU, and implement different sample selection criteria. For brevity, all regression results are summarized in Table 3.

4.3.1. Alternative measures for trade credit provision

In the baseline regression, trade credit provision is scaled by total assets. Following Love et al. (2007) and Luo et al. (2023), we replace the dependent variables using operating costs or sales as denominators instead of total assets to measure trade credit. Moreover, similar to Cao et al. (2024), we also substitute the dependent variables with a net trade credit provision measure. This is calculated by subtracting accounts payable, notes payable, and prepaid expenses from the sum of accounts receivable, notes receivable, and prepaid expenses, all scaled by total assets. We then rerun the Eq. (1) using these alternative definitions. As shown in columns (1) and (3) of Table 3, the coefficients of TC_Rev_{t+1}, TC_Cor_{t+1} and $NetTC_{t+1}$ for $CCPU$ maintain their negative signs and 1% significance levels, aligning with our baseline results.

4.3.2. Alternative measure for CCPU

So far, our CCPU proxy index has been calculated based on climate policy-related news in Chinese newspapers. However, potential news censorship may affect the accuracy of this sample to some extent. To address these concerns, we use an alternative measure of CCPU. Following Wang et al. (2026), we adopt the CCPU index constructed by Lee and Cho (2023), which is derived from climate uncertainty-related news about China posted on Twitter (now 𝕏 platform), with monthly indices calculated from 2010 to 2022. We equally weight these monthly indices to obtain our alternative CCPU variable ($CCPU_\mathbb{X}$). Column (4) of Table 3 presents the re-estimated baseline regression results using $CCPU_\mathbb{X}$, confirming that our main findings remain intact.

4.3.3. Excluding the impact of COVID-19

The COVID-19 pandemic has profoundly affected global economies, resulting in dramatic changes in policy environments and market conditions. For China, in particular, the pandemic has caused unprecedented economic volatility and uncertainty, potentially obscuring or distorting the true effect of CPU on corporate trade credit provision. To mitigate the influence of major events on our estimation results, we follow Dai and Zhang (2023) and test the robustness of our primary findings by excluding samples from 2020 to 2022. Table 3 demonstrates that the negative relationship between $CCPU$ and trade credit provision remains statistically significant at the 1% level. These findings indicate that imposing the above sample criteria does not change our primary findings.

Table 2
The effect of CCPU on trade credit provision.

Variables	(1)	(2)	(3)
	TC_{t+1}	TC_{t+1}	TC_{t+1}
CCPU	−0.0243***	−0.0118***	−0.0089***
	(0.0015)	(0.0011)	(0.0013)
EPU		−0.0042***	−0.0030***
		(0.0002)	(0.0003)
Size			−0.0047**
			(0.0020)
Age			−0.0038**
			(0.0016)
Lev			0.0379***
			(0.0066)
ROA			0.0589***
			(0.0154)
Growth			0.0008***
			(0.0002)
Cash			−0.1020***
			(0.0082)
Fix			−0.0344***
			(0.0085)
BM			−0.0061**
			(0.0030)
EBIT			−0.0005
			(0.0014)
Top1			−0.0001
			(0.0001)
Indep			−0.0041
			(0.0136)
Constant	0.2268***	0.2109***	0.3139***
	(0.0040)	(0.0034)	(0.0411)
Firm FE	Yes	Yes	Yes
N	31 924	31 924	31 924
adj. R^2	0.7901	0.7955	0.7998

Notes: This table presents the baseline regression results examining the effect of Chinese climate policy uncertainty on firms' trade credit provision. The dependent variable is one of the trade credit measures, TC. The key explanatory variable $CCPU$ represents proxies for climate policy uncertainty in China. All variables are defined in detail in Table A.1. Firm FE denotes firm fixed effects. Robust standard errors in parentheses are clustered by firms. *,**, and *** indicate significance at 10%, 5%, and 1% significance levels, respectively.

4.4. Endogeneity

The baseline regression results may be subject to endogeneity concerns. First, although this study treats CCPU as exogenous to individual micro-enterprises, government climate policy adjustments are often based on these enterprises' overall performance. Reduced trade credit provision may indicate deteriorating financial conditions, impeding firms' ability to meet existing climate policy requirements. Consequently, enterprises might lobby for government to ease or modify current climate policies. To address widespread financial challenges, the government may frequently adjust climate-related policies, balancing economic growth and environmental protection goals. Such frequent policy changes can increase CPU, potentially creating reverse causality issues.

Second, while we have controlled for individual fixed effects, climate policy implementation may vary across industries and cities. This variation raises concerns about omitted variables, presenting another potential source of endogeneity.

Third, systematic differences in firm characteristics between high and low CCPU periods may introduce selection bias. Specifically, high CCPU periods often coincide with sector-specific regulatory pressures and complex policy environments that influence firms' risk preferences and liquidity management strategies, potentially confounding our main results. We therefore must ensure that our findings are not driven by variations in observable firm characteristics across these periods.

In this section, we address identification concerns by employing three sets of empirical strategies.

4.4.1. Instrumental variable approach

We first employ an instrumental variable approach to mitigate the reverse causality concern. Following Mo and Liu (2023), our instrument is the degree of CPU in the U.S. (iv_CPU), developed by Gavriilidis (2021) based on news coverage in eight prestigious U.S. newspapers capturing U.S. climate policy changes. We argue that while U.S. climate policy developments may shape China's policy formulation, they are unlikely to directly affect Chinese firms' trade credit provision decisions.

The underlying assumption is that climate policy coordination occurs through international frameworks such as the Paris Agreement, where major economies align their climate strategies (Keohane and Victor, 2016). As the world's second-largest

Table 3
Robustness checks.

Variables	Alternative TC			Alternative CPU	Subsample regression
	(1)	(2)	(3)	(4)	(5)
	TC_Rev_{t+1}	TC_Cor_{t+1}	$NetTC_{t+1}$	TC_{t+1}	TC_{t+1}
CCPU	−0.0184***	−0.0247***	−0.0046**		−0.0056***
	(0.0033)	(0.0061)	(0.0023)		(0.0017)
CCPU_𝕏				−0.0329***	
				(0.0042)	
EPU	−0.0095***	−0.0143***	0.0017***	−0.0030***	−0.0035***
	(0.0007)	(0.0013)	(0.0005)	(0.0003)	(0.0003)
Size	0.0194***	0.0349***	−0.0049	−0.0050**	−0.0046**
	(0.0053)	(0.0099)	(0.0030)	(0.0020)	(0.0021)
Age	0.0036	−0.0183**	−0.0112***	−0.0029*	−0.0012
	(0.0044)	(0.0086)	(0.0027)	(0.0016)	(0.0017)
Lev	0.0968***	0.2496***	0.7317***	0.0389***	0.0329***
	(0.0178)	(0.0366)	(0.0117)	(0.0066)	(0.0070)
ROA	0.0044	0.1120	0.0217	0.0568***	0.0544***
	(0.0426)	(0.0851)	(0.0327)	(0.0154)	(0.0164)
Growth	−0.0005	0.0005	0.0015***	0.0007***	0.0008***
	(0.0006)	(0.0011)	(0.0004)	(0.0002)	(0.0002)
Cash	−0.3016***	−0.3758***	−0.0724***	−0.1028***	−0.1006***
	(0.0233)	(0.0434)	(0.0151)	(0.0082)	(0.0085)
Fix	−0.1531***	−0.3082***	0.0043	−0.0354***	−0.0315***
	(0.0248)	(0.0482)	(0.0138)	(0.0085)	(0.0087)
BM	−0.0163**	−0.0551***	−0.0108**	−0.0074**	−0.0058*
	(0.0078)	(0.0147)	(0.0054)	(0.0029)	(0.0030)
EBIT	−0.0140***	−0.0131**	0.0139***	−0.0001	−0.0001
	(0.0034)	(0.0064)	(0.0026)	(0.0014)	(0.0015)
Top1	−0.0002	−0.0004	0.0001	−0.0001	−0.0002
	(0.0003)	(0.0006)	(0.0002)	(0.0001)	(0.0001)
Indep	−0.0238	−0.0091	0.0284	−0.0046	−0.0018
	(0.0389)	(0.0753)	(0.0244)	(0.0136)	(0.0143)
Constant	0.1008	0.1487	0.0456	0.3012***	0.3017***
	(0.1095)	(0.2044)	(0.0641)	(0.0416)	(0.0441)
Firm FE	Yes	Yes	Yes	Yes	Yes
N	31 924	31 924	31 924	31 924	27 831
adj. R^2	0.7330	0.7323	0.8573	0.8000	0.8072

Notes: This table presents robustness check results. Columns (1) - (3) report regression results using alternative measures of trade credit provision, as defined in Section 3.2. Column (4) uses $CCPU_𝕏$, an alternative measure of CCPU developed by Lee and Cho (2023). Column (5) presents subsample regression results excluding samples from 2020–2022 to mitigate COVID-19 pandemic impacts. All regressions include identical control variables as specified in Table 2. Firm fixed effects are denoted by Firm FE. Robust standard errors, clustered by firms, are reported in parentheses. *,**, and *** indicate significance at 10%, 5%, and 1% significance levels, respectively.

economy, China closely monitors U.S. climate policy developments to maintain international commitments and competitive positioning. Moreover, U.S. climate policies often establish global standards that serve as benchmarks for Chinese policymakers, particularly in areas such as emissions standards and carbon pricing mechanisms (Zhang, 2007; Chen et al., 2018).

Regarding the exclusion restriction, our sample consists entirely of Chinese listed firms operating under domestic regulatory and institutional frameworks. In contrast, U.S. CPU reflects uncertainty within the U.S. institutional context, shaped by legal and regulatory structures that differ fundamentally from those in China. Furthermore, Chinese firms' trade credit decisions are primarily driven by domestic market dynamics – including customer relationships, creditworthiness assessments, and collection practices – which are embedded in Chinese commercial law and business customs. Therefore, any potential impact of U.S. policies on Chinese firms would operate through changes in China's own CPU rather than direct channels. Thus, it serves as a suitable instrumental variable, capturing exogenous variation in CCPU that is plausibly orthogonal to Chinese firms' credit allocation decisions.

Table 4, Columns (1) and (2), present the instrumental variable regression results. In the first stage, $CCPU$ is regressed on the instrumental variable, alongside all controls and fixed effects from the baseline regression. The iv_CPU coefficient is positive and significant at the 1% level, showing strong correlation with the $CCPU$ proxy. Furthermore, the F-statistics from the weak instrument test exceed 10, satisfying the relevance condition of the instrumental variable approach. The second-stage results show a significantly negative $CCPU$ coefficient, indicating that CPU substantially reduces firms' trade credit supply, consistent with our baseline findings.

4.4.2. Omitted variable bias

To further mitigate concerns about omitted variables on trade credit provision, our second set of tests introduces city fixed effects in Eq. (1). Likewise, we control for time trend variables to capture the overall time trend of our sample period. Specifically, we include a firm-specific time trend variable that equals 1 for a firm's first year in our sample and increases by one for each subsequent

year, helping control for unobserved time-varying factors at the firm level. The regression results, shown in Column (3) of Table 4, confirming the robust negative relationship between $CCPU$ and trade credit provision.

4.4.3. Entropy balancing

As our third approach to address potential selection bias in our baseline regression results, we employ entropy balancing as an additional endogeneity test (Cefalu et al., 2020; Jannati, 2024). We define firm-year observations with above-median CCPU as the treatment group (High-CCPU) and those with below-median CCPU as the control group (Low-CCPU). Entropy balancing then assigns weights to control group observations so that their weighted covariate distribution exactly matches that of the treatment group across all baseline control variables, achieving perfect covariate balance (Hainmueller, 2012). Unlike propensity score matching, entropy balancing retains all observations while ensuring exact balance across observable firm characteristics, thereby enhancing causal inference reliability.

Panel B of Table 4 shows that entropy balancing weights equalize the weighted means of all control variables between treatment and control groups, eliminating systematic differences in firm characteristics that could confound our main findings. Using the reweighted control group, we re-estimate the baseline specification. Column (4) of Panel A in Table 4 reports results consistent with our main regression findings.

Taken together, these findings consistently demonstrate a negative relationship between CCPU and trade credit provision across all three identification strategies, supporting the robustness of our conclusions even after addressing endogeneity concerns.

4.5. Economic channels

In this section, we investigates three plausible underlying economic channels through which further CCPU reduce corporate trade credit provision. Specifically, we hypothesize that financing constrain, information opacity, and operation risk as important factors that CCPU could affects trade credit. We examine these channels by employing a double interaction approach. Overall, these empirical tests provide further support for our causal inferences regarding the negative impact of CCPU on trade credit provision.

4.5.1. The financing constraints channel

As emphasized in the hypothesis development, heightened CPU would intensify the liquidity crisis in the market (Dai and Zhang, 2023). While this impact would increase the financing needs of some firms, in turn, suppliers of trade credit will also face liquidity constraints. This may lead them to adopt conservative trade credit strategies, thereby reducing market-wide trade credit provision.

To examine how a firm's financing constrain alters the marginal effect of CCPU on its trade credit provision, we estimate the following model:

$$TC_{i,t+1} = \beta_0 + \beta_1 CCPU_t \times WW_{i,t} + \beta_2 WW_{i,t} + \beta_3 CCPU_t + \beta_4' \mathbb{M}_{i,t} + Firm_i + \varepsilon_{i,t} \tag{2}$$

where i and t index firm and year, respectively. $WW_{i,t}$ represents the Whited-Wu financial constraint index that captures firm-level financial constraints (Whited and Wu, 2006). We transform this continuous measure into a binary indicator equal to one if a firm's financial constraint level exceeds its industry-year mean, and zero otherwise. The variables $CCPU_t \times WW_{i,t}$ in Eq. (2) are our variables of interest. The interaction term's coefficient captures how the effect of CCPU on trade credit provision differs among firms with high financing constraints. We expect β_1 to be negative and statistically significant if our hypothesis holds.

The results in Table 5 reveals that the coefficient of the interaction term is −0.0034, which is significantly negative at the 1% level. This finding suggests that the negative effect of CCPU on trade credit provision is more pronounced among firms with higher financing constraints.

4.5.2. The operating risk channel

We then examine the operating risk channel. Prior studies argue that firms' incentive to provide trade credit decreases as operating risk increases (Li et al., 2020; Aral et al., 2022). Since CCU introduces additional risks to corporate production and operations, we expect CCPU to amplify this effect. That is, operationally vulnerable firms exhibit a greater tendency to reduce trade credit provision in response to CCPU.

To explore this economics channel, we examine the following regression equation:

$$TC_{i,t+1} = \beta_0 + \beta_1 CCPU_t \times Z_{i,t} + \beta_2 Z_{i,t} + \beta_3 CCPU_t + \beta_4' \mathbb{M}_{i,t} + Firm_i + \varepsilon_{i,t} \tag{3}$$

where $Z_{i,t}$ represents the Altman Z-score measuring bankruptcy risk for firm i in year t (Altman, 1968). We use this measure as a proxy for firm operating risk and transform it into a binary indicator equal to one if a firm's bankruptcy risk exceeds its industry-year mean, and zero otherwise. The parameter of interest is β_1 in Eq. (3). Since the model predicts a negative CCPU relation when there is high operating risk, we should see a negative value of β_1.

Table 5, Column (2) report the regression results for Eq. (3). The coefficient of $CCPU_t \times Z_{i,t}$ is negative and significant at the 1% level for TC_{t+1} measures. These findings suggest that CCPU inhibits trade credit provision by firms with higher operating risk, aligning with our expectations.

Table 4
Endogeneity.

Panel A: The results of endogeneity tests				
Variables	Instrumental variable approach		Omitted variable bias	Entropy balancing results
	(1)	(2)	(3)	(4)
	CPU	TC_{t+1}	TC_{t+1}	TC_{t+1}
CCPU		−0.0101***	−0.0062***	
		(0.0033)	(0.0015)	
iv_CPU	0.3338***			
	(0.0041)			
High-CCPU				−0.0027***
				(0.0006)
EPU	−0.0615***	−0.0030***	−0.0019***	−0.0028***
	(0.0010)	(0.0003)	(0.0003)	(0.0003)
Size	0.1154***	−0.0045**	−0.0106***	−0.0063***
	(0.0042)	(0.0021)	(0.0016)	(0.0019)
Age	0.0407***	−0.0034*	−0.0088***	−0.0071***
	(0.0050)	(0.0018)	(0.0016)	(0.0016)
Lev	−0.2288***	0.0376***	0.1278***	0.0419***
	(0.0199)	(0.0067)	(0.0085)	(0.0065)
ROA	−0.0080	0.0586***	0.1700***	0.0689***
	(0.0589)	(0.0154)	(0.0226)	(0.0162)
Growth	−0.0011	0.0008***	0.0007**	0.0008***
	(0.0008)	(0.0002)	(0.0003)	(0.0002)
Cash	0.0682**	−0.1023***	−0.2304***	−0.0997***
	(0.0333)	(0.0083)	(0.0130)	(0.0083)
Fix	−0.1428***	−0.0347***	−0.1351***	−0.0323***
	(0.0256)	(0.0086)	(0.0097)	(0.0085)
BM	0.0319***	−0.0059**	−0.0054	−0.0062**
	(0.0107)	(0.0030)	(0.0050)	(0.0030)
EBIT	0.0005	−0.0005	−0.0052***	−0.0015
	(0.0053)	(0.0014)	(0.0020)	(0.0014)
Top1	−0.0020***	−0.0001	0.0000	−0.0001
	(0.0003)	(0.0001)	(0.0001)	(0.0001)
Indep	0.1604***	−0.0039	−0.0167	−0.0099
	(0.0552)	(0.0137)	(0.0213)	(0.0140)
Timetrend			−0.0001	
			(0.0003)	
Kleibergen–Paap F-statistics	28 667.01			
Cragg–Donald F-statistics	3603.25			
Firm FE	Yes	Yes	Yes	Yes
City FE	No	No	Yes	Yes
N	31 924	31 924	31 924	31 924
adj. R^2	0.0632	0.0669	0.8000	0.7991

Panel B: Entropy Balancing				
Variable	Mean Treated	Mean Pre-Weight Control	Mean Post-Weight Control	Diff.
EPU	4.168	4.587	4.168	0
Size	22.18	22.11	22.18	0
Age	2.01	1.976	2.01	0
Lev	0.4112	0.4167	0.4112	0
ROA	0.04063	0.03778	0.04063	0
Growth	−0.35	−0.4432	−0.35	0
Cash	0.04746	0.04645	0.04746	0
Fix	0.2064	0.2086	0.2064	0
BM	0.5903	0.6434	0.5903	0
EBIT	0.644	0.6079	0.644	0

(*continued on next page*)

Table 4 (*continued*).

Top1	34.09	34.57	34.09	0
Indep	0.3764	0.3756	0.3764	0

Notes: This table presents results from empirical strategies addressing potential endogeneity issues in the relationship between Chinese climate policy uncertainty and trade credit provision. Panel A reports regression results across three specifications. The first two columns report two-stage least squares regression results. Columns (1) and (2) report the two-stage least squares regression results. Column (1) shows the first-stage regression, where CCPU is regressed on instrumental variables (iv_CPU) and control variables. Column (2) presents the second-stage estimation results. Column (3) incorporates city and industry fixed effects, and a linear time trend variable (*Timetrend*). Column (4) reports entropy balancing results, where High-CCPU equals one if climate policy uncertainty exceeds the sample median, zero otherwise. Panel B reports the means of control variables before and after entropy balancing, demonstrating perfect first-moment balance between treatment and control groups. All regressions include identical control variables as specified in Table 2. Firm, industry, and city fixed effects are denoted by Firm FE, Industry FE, and City FE, respectively. Robust standard errors, clustered by firms, are reported in parentheses. *,**, and *** indicate significance at 10%, 5%, and 1% significance levels, respectively.

4.5.3. The information opacity channel

We further investigate the economics channel by exploring the variation of information opacity in the effect of CCPU in reducing trade credit provision. Information transparency is fundamental to suppliers offering trade credit (Stiglitz and Weiss, 1981; Wu et al., 2014). Typically, trade credit extensions are interest-free and unsecured, exposing suppliers to heightened credit risk. Consequently, significant information asymmetry between borrowers and lenders may curtail trade credit provision. Additionally, elevated policy uncertainty tends to intensify market-wide information opacity (Gulen and Ion, 2015). Thus, we expect the negative relationship between CCPU and trade credit provision to be stronger in industries characterized by higher levels of information opacity.

The following regression model is employed to test this conjecture:

$$TC_{i,t+1} = \beta_0 + \beta_1 CCPU_t \times SYN_{i,t} + \beta_2 SYN_{i,t} + \beta_3 CCPU_t + \beta_4' \mathbb{M}_{i,t} + Firm_i + \varepsilon_{i,t} \quad (4)$$

where $SYN_{i,t}$ refer information opacity for firm i in year t, measured by stock price synchronicity following Liu and Hou (2019). Stock price synchronicity captures the extent to which individual stock prices move together with market and industry indices, with higher synchronicity indicating that firm-specific information is less incorporated into stock prices (Morck et al., 2000; Jin and Myers, 2006). This measure is directly reflects the availability and incorporation of firm-specific information in financial markets, which is crucial for suppliers' credit assessment decisions. We transform this measure into a binary indicator that equals one when an industry's stock price synchronicity surpasses the annual sample mean, and zero otherwise. The variable of interest in Eq. (4) is the interaction term, $CCPU_t \times SYN_{i,t}$.

Column (3) of Table 5 reports estimation results. As expected, we observe a significantly negative coefficient on the interaction term, $CCPU_t \times SYN_{i,t}$, at the 1% level. This finding suggests that $CCPU$ has a more pronounced negative effect on trade credit provision for firms with low information transparency.

4.6. Cross-sectional heterogeneity

In this subsection, we conduct cross-sectional tests that make use of variation in several characteristics of China's unique enterprise nature of property rights (i.e., state-owned and non-state-owned enterprises), informal institutions (i.e., political connections), and the product market competition environment in which firms operate to provide more additional evidence supporting our main argument regarding the effect of CCPU on trade credit provision.

4.6.1. The nature of property rights

We first examine the differences between SOEs and Non-SOEs under CCPU. SOEs are a crucial component of the Chinese economy. Cull et al. (2009) and Chen et al. (2021) document that state ownership positively correlates with trade credit provision in China, where financial markets remain underdeveloped. This is because SOEs, compared to Non-SOEs, typically have more abundant financial resources and stronger government support, enabling them to extend more trade credit to the market. Furthermore, Cao et al. (2024) argue that periods of high CPU lead to more constrained external financing conditions. We therefore expect that this may result in smaller trade credit provision by Non-SOEs compared to SOEs.

To examine this conjecture, we performed subsample regressions based on firms ownership structure. The result are shown in column (1) - (2) of Table 6. The CCPU coefficient for Non-SOEs is −0.0110, significant at the 1% level, while for SOEs, the estimated coefficient of CCPU is insignificant. These findings indicate that the negative effect of CCPU on trade credit provision is more pronounced in Non-SOEs, which aligns with our expectations.

4.6.2. The political connections

We next examine how political connections moderate the relationship established in the Eq. (1). Prior studies argue that politically connected firms possess greater informational and resource allocation advantages to refrain from policy uncertainty that cause negative externalities, thereby maintaining corporate value stability (Boubakri et al., 2012; Liu et al., 2021). These connections can also bring potential benefits to companies, including lower regulatory requirements (Goldman et al., 2008), lighter taxation (Faccio, 2010), and reduced financing costs (Sohail Mansha and Hooy, 2022). Consequently, the reduction in trade credit provision attributable to CCPU that we document above may be less pronounced in firms with higher political connections.

Table 5
The regression result of economic channels.

Variables	Financing constraints channel	Operating risk channel	Information opacity channel
	(1)	(2)	(3)
	TC_{t+1}	TC_{t+1}	TC_{t+1}
CCPU × WW	−0.0035***		
	(0.0006)		
WW	0.0068***		
	(0.0015)		
CCPU × Z		−0.0024***	
		(0.0006)	
Z		−0.0000	
		(0.0016)	
CCPU × SYN			−0.0020***
			(0.0006)
SYN			0.0034***
			(0.0010)
CCPU	−0.0051***	−0.0055***	−0.0070***
	(0.0014)	(0.0015)	(0.0013)
EPU	−0.0029***	−0.0030***	−0.0029***
	(0.0003)	(0.0003)	(0.0003)
Size	−0.0050**	−0.0046**	−0.0047**
	(0.0020)	(0.0020)	(0.0020)
Age	−0.0035**	−0.0034**	−0.0037**
	(0.0016)	(0.0016)	(0.0016)
Lev	0.0373***	0.0410***	0.0382***
	(0.0066)	(0.0069)	(0.0066)
ROA	0.0532***	0.0573***	0.0576***
	(0.0153)	(0.0154)	(0.0154)
Growth	0.0007***	0.0008***	0.0008***
	(0.0002)	(0.0002)	(0.0002)
Cash	−0.1043***	−0.1023***	−0.1022***
	(0.0083)	(0.0082)	(0.0082)
Fix	−0.0345***	−0.0347***	−0.0349***
	(0.0085)	(0.0085)	(0.0085)
BM	−0.0070**	−0.0050	−0.0066**
	(0.0030)	(0.0031)	(0.0030)
EBIT	0.0001	−0.0005	−0.0003
	(0.0014)	(0.0014)	(0.0014)
Top1	−0.0001	−0.0001	−0.0001
	(0.0001)	(0.0001)	(0.0001)
Indep	−0.0030	−0.0044	−0.0039
	(0.0136)	(0.0136)	(0.0136)
Constant	0.3087***	0.3021***	0.3081***
	(0.0424)	(0.0415)	(0.0415)
Firm FE	Yes	Yes	Yes
N	31 924	31 924	31 924
adj. R^2	0.8002	0.8000	0.7999

Notes: This table presents results on the impact of CCPU on trade credit provision through various economic channels. Columns (1)-(3) test different channels using binary indicators based on sample means. Column (1) uses the WW index (Whited and Wu, 2006) for financing constraints. Column (2) employs the Z-score (Altman, 1968) for operating risk. Column (3) applies the SYN index (Liu and Hou, 2019) for information opacity. For each measure, the indicator equals one when the firm's value exceeds the annual sample mean, zero otherwise. All regressions include identical control variables as specified in Table 2. Firm fixed effects are denoted by Firm FE. Robust standard errors, clustered by firms, are reported in parentheses. *,**, and *** indicate significance at 10%, 5%, and 1% significance levels, respectively.

To test this prediction, we refer to Cheng et al. (2017) and Liu et al. (2021), identify politically connected firms as those with top executives who either served as government officials or are members of the Chinese Communist Party. We construct binary indicator, PC, which equals one if an executive meets either of these criteria, and zero otherwise. We re-estimate Eq. (1) and present the results in columns (3) and (4) of Table 6. For firms with low political connections, the estimated coefficient of CCPU is −0.0118, significant at the 1% level. In contrast, firms with high political connections show a negative coefficient of CCPU that is only significant at the 10% level. These findings align with our expectation that political connections moderate the negative effect of CCPU on trade credit provision.

4.6.3. The product market competition

We also consider whether the effect of CPU varies with product market competition. According to Dhaliwal et al. (2014), the external competitive environment impacts firms' financial conservatism. Giannetti et al. (2021) provides direct evidence that suppliers extend trade credit to customers with high bargaining power when they fear sales cannibalization by other customers. That is, intense industry competition increases firms' propensity to both extend and receive trade credit financing. Thus, to the

Table 6
Cross-sectional heterogeneity.

Variables	Non-SOEs	SOEs	Low-PC	High-PC	Low-Competition	High-Competition
	(1)	(2)	(3)	(4)	(5)	(6)
	TC_{t+1}	TC_{t+1}	TC_{t+1}	TC_{t+1}	TC_{t+1}	TC_{t+1}
CCPU	−0.0110***	−0.0022	−0.0118***	−0.0028*	−0.0131***	−0.0025
	(0.0016)	(0.0020)	(0.0020)	(0.0016)	(0.0021)	(0.0019)
EPU	−0.0024***	−0.0037***	−0.0022***	−0.0033***	−0.0030***	−0.0026***
	(0.0004)	(0.0004)	(0.0005)	(0.0004)	(0.0003)	(0.0004)
Size	−0.0076***	−0.0032	−0.0014	−0.0066**	−0.0040*	−0.0036
	(0.0027)	(0.0032)	(0.0033)	(0.0026)	(0.0023)	(0.0026)
Age	−0.0022	−0.0050	−0.0036	−0.0056**	−0.0028	−0.0053**
	(0.0019)	(0.0034)	(0.0024)	(0.0023)	(0.0021)	(0.0022)
Lev	0.0297***	0.0458***	0.0247**	0.0402***	0.0333***	0.0396***
	(0.0083)	(0.0119)	(0.0106)	(0.0090)	(0.0088)	(0.0078)
ROA	0.0552***	0.0586**	0.0645***	0.0662***	0.0639***	0.0493**
	(0.0186)	(0.0295)	(0.0224)	(0.0218)	(0.0216)	(0.0209)
Growth	0.0008***	0.0007**	0.0009**	0.0006**	0.0008***	0.0007**
	(0.0003)	(0.0003)	(0.0004)	(0.0003)	(0.0003)	(0.0003)
Cash	−0.1274***	−0.0649***	−0.1236***	−0.0807***	−0.1171***	−0.0887***
	(0.0109)	(0.0126)	(0.0138)	(0.0100)	(0.0111)	(0.0127)
Fix	−0.0192*	−0.0520***	−0.0047	−0.0459***	−0.0364***	−0.0238**
	(0.0109)	(0.0140)	(0.0142)	(0.0113)	(0.0108)	(0.0099)
BM	−0.0056	−0.0086*	−0.0049	−0.0107***	−0.0060	−0.0093**
	(0.0037)	(0.0051)	(0.0046)	(0.0039)	(0.0039)	(0.0041)
EBIT	0.0019	−0.0046**	−0.0015	−0.0017	−0.0005	0.0002
	(0.0018)	(0.0022)	(0.0021)	(0.0018)	(0.0018)	(0.0019)
Top1	−0.0002	−0.0001	−0.0001	−0.0001	−0.0002*	−0.0000
	(0.0002)	(0.0002)	(0.0002)	(0.0001)	(0.0001)	(0.0001)
Indep	−0.0051	0.0036	−0.0341	0.0094	0.0039	−0.0111
	(0.0179)	(0.0205)	(0.0229)	(0.0172)	(0.0175)	(0.0183)
Constant	0.3916***	0.2499***	0.2675***	0.3365***	0.3180***	0.2645***
	(0.0567)	(0.0671)	(0.0697)	(0.0560)	(0.0490)	(0.0545)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
N	20 288	11 617	13 899	17 954	16 092	14 967
adj. R^2	0.7745	0.8302	0.7765	0.8257	0.7911	0.7978

Notes: This table examines how the impact of CCPU on trade credit provision varies across different groups based on property rights, political connections, and product market competition. Columns (1)–(2) compare "Non-SOEs" versus "SOEs". Columns (3)–(4) contrast firms with low versus high political connections (*PC*), based on Cheng et al. (2017) and Liu et al. (2021) (see Table A.1 for details). Columns (5)–(6) compare industries with low versus high competition levels (Jory et al., 2020). All regressions include identical control variables as specified in Table 2. Firm fixed effects are denoted by Firm FE. Robust standard errors, clustered by firms, are reported in parentheses. *, **, and *** indicate significance at 10%, 5%, and 1% significance levels, respectively.

extent that CPU impose this negative effect on firms' trade credit provision, we expect this relationship to be more pronounced in environments with low product market competition.

To capture this effect more precisely, we employ the Herfindahl–Hirschman Index (HHI) following Jory et al. (2020) to measure product market competition, where lower HHI values indicate higher competitive intensity. We partition our sample based on the industry-year median HHI and report results in columns (5) and (6) of Table 6. The findings reveal that the effect of CCPU is negative but statistically insignificant in high-competition markets, while exhibiting a significant negative effect at the 1% level in low-competition environments. These findings demonstrate that product market competition mitigates the negative effect of CCPU on trade credit provision, aligning with our expectations.

5. Conclusions

Using panel data from China A-share listed firms during 2010–2022, we investigate how CCPU affects corporate trade credit provision. We hypothesize that CCPU disrupts market and business operations, thereby reducing firms' trade credit provision. Supporting this conjecture, our empirical results confirm a significant negative correlation between CCPU and firms' trade credit provision. This finding remains robust across alternative measurements of our variable of interest, utilizing subsamples, applying instrumental variables approaches, and incorporating additional fixed effects. We then identify three economic channels underlying the negative effect of CCPU on trade credit: the financing constraint channel, the operating risk channel, and the information opacity channel. Furthermore, our cross-sectional tests support the argument that this effect is more pronounced in non-state-owned enterprises, industries with weaker competition, and firms with fewer political connections.

Our findings have broader relevance beyond China, particularly in light of the evolving global climate policy landscape following the 2015 Paris Agreement. This international accord, signed by 195 countries, established a coordinated framework requiring extensive policy adjustments to meet Nationally Determined Contributions, thereby generating widespread CPU across developed and emerging economies. While our analysis focuses on China, several findings likely extend to other institutional contexts. The three

Table A.1
Variable definitions.

Symbol	Measurements
TC	(Accounts Receivable + Notes Receivable + Prepayments)/Total Assets.
TC_Rev	(Accounts Receivable + Notes Receivable + Prepayments)/Sales.
TC_Cor	(Accounts Receivable + Notes Receivable + Prepayments)/Operating costs.
NetTC	(Accounts Receivable + Notes Receivable + Prepaid Accounts) - (Accounts Payable + Notes Payable + Prepaid Accounts)/Total Assets.
CCPU	Climate policy uncertainty in China. Index, see https://figshare.com/articles/dataset/ChinasCPUindex/24071193/1.
CCPU_𝕏	An alternative measure of Chinese climate policy uncertainty. Index, see https://twitterchnepu.github.io/.
EPU	Policy uncertainty in China. Index, see http://www.policyuncertainty.com/china_epu.html.
Size	Natural logarithm of total assets.
Age	Natural logarithm of the number of years since the firm was listed on the exchange.
Lev	The ratio of the firm's total liabilities to total assets at the end of the fiscal year.
ROA	Total net earnings over total assets.
Growth	The annual growth rate of sales.
Cash	Operating cash flow divided by total assets.
BM	Book-to-market value ratio.
EBIT	Total earnings before tax and interest over total assets.
Top1	The proportion of shares owned by the largest shareholder.
Indep	The number of independent directors divided by the total number of directors.
iv_CPU	Climate Policy Uncertainty in the U.S.. Index, see https://www.policyuncertainty.com/climate_uncertainty.html.
WW	A binary indicator equal to one if a firms' financial constraint level, measured by the WW index, exceeds the industry-year mean, and zero otherwise. The WW index is calculated as: $WW = -0.091 * CF - 0.062 * DivPos + 0.021 * Lev - 0.044 * Size + 0.102 * ISG - 0.035 * SG$ where CF represents the ratio of operating cash flow to total assets. $DivPos$ is a binary indicator that equals one for dividend-paying firms. Lev denotes the ratio of long-term debt to total assets. $Size$ is calculated as the natural logarithm of total assets. ISG represents the industry sales growth rate, and SG denotes the firm-specific sales growth rate.
Z	A binary indicator equal to one if a firms' operating risk level, measured by the Z index, exceeds the industry-year mean, and zero otherwise. The Z index is calculated as: $Z = 1.2 * \frac{WC}{TA} + 1.4\frac{retEarnings}{TA} + 3.3 * \frac{EBIT}{TA} + 0.6 * \frac{MV}{TA} + 0.999 * \frac{SALE}{TA}$ where WC represents working capital; TA denotes total assets; $retEarnings$ represents retained earnings; $EBIT$ refers to earnings before interest and taxes; MV denotes the market value of equity; TL represents total liabilities; $Sale$ refers to sales.
SYN	A binary indicator equal to one if a firm's information opacity level, measured by the SYN index, exceeds the industry-year mean, and zero otherwise. The SYN index is calculated as: $SYN_i = ln(\frac{R_i^2}{1-R_i^2})$ $R_{i,t} = \alpha_0 + \alpha_1 R_{m,t} + \alpha_2 R_{I,t} + \varepsilon_{i,t}$ where $R_{i,t}$ represents the rate of return of stock i in week t, including reinvested cash dividends. $R_{m,t}$ denotes the market capitalization-weighted average return of all A-share stocks in week t. $R_{I,t}$ represents the market value-weighted average return, calculated using each firm's circulating market value as weights. A higher index value indicates a greater degree of information asymmetry.
SOE	A binary indicator equals to one if a firm is a state owned enterprise, and zero otherwise.
PC	A binary indicator that equals one for firms whose top executives have either governmental experience or Chinese Communist Party membership, and zero otherwise.
HHI	A binary indicator equal to one if a firms' industry product market competition level, measured by the HHI index, exceeds the market-year mean, and zero otherwise. The HHI index is calculated as: $HHI = \sum_i^N (X_i/X)^2$ where X represents the total industry size, X_i denotes the size of firm i, and N represents the number of firms in the industry. This index is negatively correlated with industry competition intensity. The index is transformed into a binary variable: industries with values above the market average are assigned 0, while those below are assigned 1.

channels through which CPU influences trade credit provision—financing constraints, operational risks, and information opacity—represent fundamental economic mechanisms that transcend national boundaries. These transmission channels operate in similar ways across economies experiencing CPU, particularly in emerging markets characterized by bank-dominated financial systems, substantial government intervention, and heavy reliance on alternative financing sources. However, the magnitude of these effects may vary depending on local financial market development and institutional characteristics.

These findings hold significant implications for various stakeholders alike. From a practical perspective, our results suggest that the substantial impact of CPU on trade credit provision emphasizes the importance of adopting stable corporate financing policies during periods of heightened uncertainty, particularly in economies where trade credit plays a crucial role in supplementing formal bank financing. Specific measures may include establishing credit guarantee schemes or providing market liquidity support. Meanwhile, when governments design and adjust environmental policies to meet Paris Agreement requirements, they should fully account for the negative effects of policy uncertainty. Clear policy communication, phased implementation schedules, and harmonized disclosure standards can help minimize disruptions in trade credit markets caused by uncertainty.

For corporate managers, our findings underscore the importance of proactive risk management in addressing CPU. Firms can mitigate liquidity shocks by diversifying financing sources—such as building relationships with multiple banks, accessing capital markets, or securing credit facilities from non-bank institutions. Supply chain financing stability can be improved through long-term contracts with key suppliers and customers, which reduce dependence on short-term trade credit. In addition, firms should strengthen information systems to minimize asymmetries with trading partners. Companies in climate-sensitive industries should also integrate climate policy scenarios into trade credit risk assessments, especially when dealing with financially constrained counterparts.

Overall, our research, by identifying CCPU as a significant determinant of trade credit provision, contributes new evidence to the trade credit literature and offers valuable insights for government policymakers and corporate management. We suggest that it would be useful for Chinese policymakers to establish clear directions and timely disclosure standards for climate policy. This approach could help enterprises respond to policy changes proactively, thus mitigating the social impact of uncertainty. Furthermore, we also suggest that corporate management accurately assess the negative effect of CPU on trade credit and implement effective risk management strategies to ensure stable company operations.

CRediT authorship contribution statement

Zihao Ning: Writing – review & editing, Writing – original draft, Formal analysis, Software, Methodology, Data curation, Conceptualization, Supervision. **Zhibo Xu:** Writing – review & editing, Writing – original draft.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix. Variable definitions

See Table A.1.

Data availability

Data will be made available on request.

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